

GRAFT: Gradient-Field Representations and Frozen-Backbone Fusion for Robust Detection of AI-Generated Images

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WORKING DRAFT. Method, protocol, and related work are complete; all experimental numbers are pending and marked **[TO FILL:]**. This document must not be submitted or distributed as a completed study until every placeholder is replaced with results reproduced by the released code.

Abstract

Detectors of AI-generated images fail along two axes that matter in deployment: they collapse on images from *unseen* generators, and they collapse when images traverse real transport chains (JPEG re-encoding, resizing, platform re-processing). We present GRAFT, a detection architecture that fuses two families of evidence with complementary failure modes: a *learnable gradient-field branch* operating on native-resolution crops — combining steganalysis-style high-pass residuals, an up-sampling (NPR-style) residual, and, novel in this context, the *structure-tensor coherence* of the image gradient field — with *frozen* vision-language-model features, whose resistance to generator overfitting is preserved precisely because the backbone is never fine-tuned. A gated fusion head with per-branch deep supervision yields a single temperature-calibrated probability together with interpretable per-branch scores, and an explicit abstention band. Training passes every image through a simulated transport chain, making robustness a training-time property rather than an evaluation afterthought. Under the strictest GenImage protocol (train on Stable Diffusion v1.4 only, test on eight generators), GRAFT attains **[TO FILL: mean AUROC]**, retaining **[TO FILL: AUROC at JPEG quality 50]** under recompression, versus **[TO FILL: best baseline]**. Code, weights, and the full evaluation harness are released.

1 Introduction

Photorealistic image generators — latent diffusion systems, autoregressive image models, and their commercial descendants — have made fabricated imagery cheap at consumer scale. The forensic response, a decade of increasingly capable “real vs. generated” classifiers, suffers from a well-documented pair of gaps between benchmark and deployment. First, *generator generalization*: a detector trained

on one generator family tends to key on that family’s idiosyncratic artifacts and degrades sharply on newer models [17, 6, 20]. Second, *transport robustness*: the high-frequency traces most detectors exploit are attenuated or destroyed by the compression-resize pipelines of messaging and social platforms [6, 1], and adversarial post-processing can defeat detectors with theoretical backing [14].

The literature contains two strong but largely separate responses. One line makes *low-level traces* more generator-agnostic: gradients of the image with respect to a pretrained model (LGrad [16]), local up-sampling residuals (NPR [15]), and spectral statistics [3, 4]. A second line abandons low-level traces altogether and reads *frozen foundation-model feature spaces*, in which generated images cluster even for unseen generators (UniversalFakeDetect [10]; CLIP-based successors [2, 9]). These families fail differently: low-level branches are strongest on clean, native-resolution inputs and degrade under recompression; frozen-feature branches survive transport better but are weaker at separating high-fidelity images whose semantics are unremarkable. Hybrid designs are beginning to appear [19], but — to our knowledge, as of this draft — no published method fuses a *learnable gradient-field representation* with a *frozen* vision-language backbone under a *robustness-first* training protocol with calibrated, abstaining outputs. GRAFT is that design, released as a maintained open-source system rather than a one-off research artifact.

Our contributions:

1. **A gradient-field forensic representation** stacking SRM high-pass residuals [5], an NPR-style up-sampling residual [15], gradient magnitude, and — new in this context — the structure-tensor *coherence* of the gradient field, an explicit, resolution-preserving measure of the anomalous local gradient alignment of synthetic texture (§3.1).
2. **Gated frozen-backbone fusion** with per-branch deep supervision: complementary evidence is combined by a learned gate whose weights, together with per-branch probabilities, are exposed as an interpretability signal; the semantic backbone stays frozen to preserve its cross-generator behavior (§3.4).

3. **A robustness-first protocol:** transport-chain augmentation at training time; native-resolution multi-crop inference that never resamples the evidence; temperature-calibrated probabilities with an abstention band; and mandatory reporting of perturbed-set metrics, TPR at fixed low FPR, and calibration error alongside clean AUROC (§3.5).
4. **Reproducible release:** training, evaluation, and robustness harnesses, plus weights, as the open-source system `fakeradar`.

2 Related Work

CNN classifiers and their limits. Wang et al. [17] showed a ResNet trained on one GAN transfers surprisingly well across CNN generators given aggressive augmentation — and established the now-standard caveat that this transfer is fragile. Gragnaniello et al. [6] documented how quickly such detectors decay under resizing and compression; Corvi et al. [1] extended the critical analysis to diffusion models.

Gradient- and residual-domain detection. Steganalysis rich models [5] established fixed high-pass residuals as content-suppressing noise descriptors. LGrad [16] converts images into gradients of a pretrained CNN, learning on a generalized artifact representation; NPR [15] shows that a simple neighboring-pixel relation targeting up-sampling operators generalizes broadly across CNN-based generators. GRAFT’s gradient branch is a superset of these signals and adds the structure-tensor coherence channel; unlike LGrad/NPR it is trained *jointly* with, and gated against, frozen semantic features.

Frequency-domain detection. Up-convolutions imprint spectral anomalies [3], and frequency inputs alone support strong detection [4]. We retain an azimuthally averaged log-power spectrum as a compact, cheap branch rather than a stand-alone detector.

Frozen foundation-model features. Ojha et al. [10] demonstrated that nearest-neighbor and linear probes in frozen CLIP [12] space generalize across generator families where end-to-end CNNs fail; Cozzolino et al. [2] and FatFormer [9] strengthen this line. We deliberately keep this branch frozen and lightweight (linear projection only), treating it as ballast for distribution shift.

Reconstruction-error methods. DIRE [18] and AEROBLADE [13] detect diffusion images via reconstruction error under a diffusion model or its autoencoder. These require running (parts of) a generator at inference; we exclude them from GRAFT’s core to preserve local, CPU-class deployment, and treat them as an optional plug-in branch in the released system.

Benchmarks and robustness limits. GenImage [20] standardizes the cross-generator protocol we adopt; Chameleon [19] stresses in-the-wild difficulty and its companion method AIDE demonstrates mixture-of-experts hybrids; Community Forensics [11] scales generator diversity in training. Saberi et al. [14] give fundamental limits and practical attacks; we therefore report perturbed metrics as first-class results and position GRAFT’s claims within, not against, those limits.

3 Method

Given image $x \in [0, 1]^{3 \times H \times W}$, GRAFT outputs $\hat{p} = P(y=\text{ai} \mid x)$, per-branch probabilities, and gate weights. Branches consume *native-resolution* crops $c \in [0, 1]^{3 \times S \times S}$ (default $S=256$); only the semantic branch sees a resized global view. Nothing the forensic branches consume is ever resampled (upsampling occurs only if $\min(H, W) < S$).

3.1 Gradient-field branch

The branch input is a 17-channel stack $\Phi(c)$ of gradient-domain maps.

(i) SRM residuals. Four fixed 5×5 high-pass kernels $\{w_k\}_{k=1}^4$ (the KV kernel, a Laplacian, and Scharr-normalized horizontal/vertical derivative kernels), applied depthwise: $r_{ch}^{(k)} = w_k * c_{ch}$, giving 12 channels that suppress content and retain synthesis noise [5].

(ii) Up-sampling (NPR-style) residual. With bilinear operators $D_{\downarrow 2}, U_{\uparrow 2}$,

$$r_{\text{npr}} = c - U_{\uparrow 2}(D_{\downarrow 2}(c)), \quad (1)$$

isolating traces of the up-sampling stages ubiquitous in generative decoders [15] (3 channels).

(iii) Gradient magnitude and coherence. Let I be the luma of c and $g = (g_x, g_y) = (\text{Sch}_x * I, \text{Sch}_y * I)$ its Scharr gradient field. With a 7×7 box filter B we form the structure tensor $J = B * (gg^\top)$ and its *coherence*

$$\kappa = \frac{\sqrt{(J_{11} - J_{22})^2 + 4J_{12}^2}}{J_{11} + J_{22} + \epsilon} \in [0, 1], \quad (2)$$

the normalized eigenvalue anisotropy of local gradient orientation. Camera imagery exhibits characteristic coherence statistics induced by optics, demosaicing, and natural texture; synthesized micro-texture — produced by up-sampling stacks and attention-mediated decoders — exhibits anomalous local gradient alignment. To our knowledge, coherence of the gradient field has not previously been used as an explicit input representation for generated-image detection; we ablate its contribution in §5.1. Together with $|g|$ this adds 2 channels.

A ResNet-18 [8] encoder (first convolution widened to 17 input channels, $\sim 11\text{M}$ parameters) maps $\Phi(c)$ to $f_{\text{grad}} \in \mathbb{R}^{512}$. Optionally, an LGrad-style channel [16] — the input-gradient of a frozen pretrained classifier — can be concatenated; it is disabled in the default configuration to keep the fast tier download-free.

3.2 Frequency branch

For luma spectrum $F = \mathcal{F}(I)$ and annuli A_ρ of normalized radius ρ , the azimuthally averaged log-power profile is

$$P(\rho) = \frac{1}{|A_\rho|} \sum_{(u,v) \in A_\rho} \log(1 + |F(u,v)|), \quad \rho \in \{1, \dots, 128\}, \quad (3)$$

standardized per image and concatenated with four band-energy statistics; a two-layer MLP yields $f_{\text{freq}} \in \mathbb{R}^{256}$ [3, 4].

3.3 Frozen semantic branch

A CLIP ViT-B/16 image encoder [12], weights frozen, maps the resized global view to features projected by a single trainable linear layer to $f_{\text{sem}} \in \mathbb{R}^{768}$. Freezing is a design requirement, not a compute convenience: fine-tuning collapses the cross-generator structure of the space [10].

3.4 Gated fusion, deep supervision, calibration

Each branch feature is projected to a shared space, $z_b = \text{GELU}(\text{LN}(W_b f_b))$, and combined by a learned gate $\alpha = \text{softmax}(\gamma) \in \Delta^{B-1}$:

$$h = \sum_b \alpha_b z_b, \quad s = \text{MLP}(h), \quad \hat{p} = \sigma(s/T), \quad (4)$$

with temperature T fitted post hoc on validation data [7]. Auxiliary heads give per-branch logits s_b ; the loss is

$$\mathcal{L} = \text{BCE}(y, s) + \frac{\lambda}{B} \sum_b \text{BCE}(y, s_b), \quad \lambda = 0.3. \quad (5)$$

Deep supervision stabilizes optimization and makes $(\alpha_b, \sigma(s_b))$ a per-image explanation of *which evidence fired*. At inference, logits are averaged over $K \leq 8$ native crops (center plus corners), and $\hat{p}$ is mapped to $\{\text{likely_real}, \text{uncertain}, \text{likely_ai}\}$ via thresholds (τ_r, τ_f) ; abstention is an explicit, reported outcome.

3.5 Robustness-first training

Every training image passes a stochastic *transport chain*: JPEG re-encoding (quality $\sim \mathcal{U}\{30, \dots, 95\}$, $p=0.5$), WebP ($p=0.1$), rescaling (factor $\mathcal{U}[0.5, 1.5]$, $p=0.3$), Gaussian blur ($p=0.15$), sensor-style noise ($p=0.1$), and horizontal flip — before random cropping. This follows the augmentation-for-generalization insight of [17], extended to the full re-encode-resize pipeline that field images actually traverse, and paired with *mandatory* perturbed-set reporting at evaluation.

Table 1: Cross-generator generalization (GenImage; train: SD v1.4 only). AUROC per test subset. **[TO FILL: complete after training runs]**

Method	Mean	Worst subset	TPR@5%FPR
CNNDetection [17]	[TO FILL:]	[TO FILL:]	[TO FILL:]
LGrad [16]	[TO FILL:]	[TO FILL:]	[TO FILL:]
NPR [15]	[TO FILL:]	[TO FILL:]	[TO FILL:]
UnivFD [10]	[TO FILL:]	[TO FILL:]	[TO FILL:]
DIRE [18]	[TO FILL:]	[TO FILL:]	[TO FILL:]
FatFormer [9]	[TO FILL:]	[TO FILL:]	[TO FILL:]
AIDE [19]	[TO FILL:]	[TO FILL:]	[TO FILL:]
GRAFT (fast)	[TO FILL:]	[TO FILL:]	[TO FILL:]
GRAFT (accurate)	[TO FILL:]	[TO FILL:]	[TO FILL:]

4 Experimental Setup

Datasets and protocol. Primary: GenImage [20] under its strictest split — train on Stable Diffusion v1.4 fakes plus ImageNet reals *only*; test on all eight subsets (ADM, BigGAN, GLIDE, Midjourney, SD v1.4, SD v1.5, VQDM, Wukong). Secondary: the in-the-wild Chameleon benchmark [19], and a held-out set of post-2025 commercial generators collected for the released system **[TO FILL: enumerate final generator list and collection dates]**.

Baselines. CNNDetection [17], LGrad [16], NPR [15], UniversalFakeDetect [10], DIRE [18], FatFormer [9], and AIDE [19], all retrained or re-evaluated under the identical protocol and identical perturbations using released code where available **[TO FILL: note any baseline that could not be reproduced]**.

Metrics. AUROC and AP (ranking quality); accuracy at 0.5; TPR at 5% FPR (the operationally relevant regime — falsely accusing real photographs is the costly error); expected calibration error (ECE); and abstention rate at $(\tau_r, \tau_f) = (0.35, 0.65)$.

Implementation. PyTorch; single NVIDIA A100 40GB; AdamW (lr 2×10^{-4} , weight decay 0.05), cosine schedule with 500-step warmup, batch 64, 10 epochs, mixed precision; crop $S=256$; the semantic backbone contributes no gradients. The fast tier (gradient + frequency) has $\sim 12\text{M}$ trainable parameters; the accurate tier adds the frozen ViT-B (+86M frozen, +0.6M trainable). Full configuration files ship with the code.

5 Results

All numbers below are pending; the tables define the exact deliverable of the experimental phase and will be populated only by runs reproducible from the released harness.

Table 2: Transport robustness: mean AUROC over the eight GenImage subsets under perturbation. **[TO FILL: complete after training runs]**

Method	clean	JPEG50	down $\times 0.5$	blur $\times 1$
Best low-level baseline	[TO FILL:]	[TO FILL:]	[TO FILL:]	[TO FILL:]
Best CLIP baseline	[TO FILL:]	[TO FILL:]	[TO FILL:]	[TO FILL:]
GRAFT (accurate)	[TO FILL:]	[TO FILL:]	[TO FILL:]	[TO FILL:]

5.1 Ablations

Planned and implemented in the harness: (a) remove each branch; (b) remove the coherence channel κ (Eq. 2) alone; (c) replace native-resolution crops with global resize to 256^2 ; (d) disable transport-chain augmentation; (e) fine-tune vs. freeze the semantic backbone; (f) gate analysis: distribution of α across clean vs. perturbed inputs; (g) calibration on/off (ECE, abstention). **[TO FILL: ablation table]**

6 Limitations and Ethics

GRAFT operates within known fundamental limits: sufficiently strong adversarial post-processing can defeat any passive detector [14], and the generator population is non-stationary — claims here are snapshots, which is why the release couples the paper to a continuously refreshed evaluation harness rather than a frozen artifact. Scores are probabilistic signals; we explicitly caution against use as sole evidence in legal, journalistic, or academic-integrity decisions, and the abstention band exists to make “the model does not know” a first-class output. Partially generated images (inpainting, AI-edited photographs) and localization are out of scope for this work. Detection research is dual-use: publishing which traces detectors read can guide evasion; we follow the field’s established norm that transparent, auditable detection — with honest robustness reporting — serves defenders more than secrecy serves attackers.

7 Conclusion

GRAFT unifies the two most generalization-robust families of forensic evidence — gradient-domain local statistics, extended with a structure-tensor coherence representation, and frozen foundation-model features — under a gated, calibrated, robustness-first protocol, released as a maintained open-source system. **[TO FILL: one-sentence summary of empirical findings once tables are complete.]**

Reproducibility. All code, configurations, manifests, and the exact benchmark and perturbation harness are released at github.com/jitendra-jha/fakeradar; weights are published with model cards on the Hugging Face Hub.

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